Chapter 1

Introduction

1.1 Background of the Study

In recent years, the field of computer vision has undergone a radical transformation, shifting from manually engineered systems to those powered by artificial intelligence. Within this domain, object detection (OD) serves as a fundamental building block for modern engineering applications, particularly in autonomous systems and national security. The evolution of aircraft detection has transitioned from traditional radar-based and optical systems toward deep learning-based frameworks capable of simultaneously solving localization and classification.

Current technological trends emphasize the use of Deep Convolutional Neural Networks (CNNs) to bridge the "semantic gap" that previously hindered image understanding. Real-world applications of these systems are now widespread, ranging from autonomous drone navigation and civil aviation safety to advanced anti-aircraft defense systems. The integration of high-speed algorithms like the You Only Look Once (YOLO) family has allowed for real-time monitoring of airspace, which is critical in scenarios where a fraction of a second can determine the success of an intercept mission.

1.2 Problem Statement

Despite significant advancements, detecting aircraft remains a formidable engineering challenge due to extreme scale variations, varying environmental lighting, and the "small object" problem. Traditional aircraft detection relies heavily on manual optical observation or expensive radar clusters, both of which have significant shortcomings. Optical methods are highly subjective and weather-dependent, while legacy radar systems often struggle with computational complexity and low-resolution output for smaller aerial targets. Existing deep learning solutions often face a trade-off between accuracy and inference speed; two-step detectors are often too slow for active defense, while one-step detectors historically struggle with high-precision localization of distant aircraft. There is a critical need for an optimized, unified framework that can provide high-fidelity detection on low-power edge devices.

1.3 Significance of the Project

This project is significant for both its academic and practical contributions:

- **Academic Importance:** It explores the limits of NMS-free inference and architectural simplification in deep learning, providing a benchmark for the latest multi-task vision frameworks.
- **Practical Relevance:** It addresses the strategic need for indigenous air defense capabilities, specifically for regions requiring cost-effective alternatives to foreign technology.
- **Expected Benefits:** The research aims to reduce the latency in the detection-to-intercept loop, enabling faster response times in autonomous defense and improved accuracy in monitoring high-density airspace.

1.4 Project Objectives

The general objective of this study is to implement and evaluate a deep learning-based system for the real-time detection and tracking of aerial targets. Specific objectives include:

- To design an efficient neural network architecture optimized for small-object aerial detection.
- To analyze the performance of various YOLO iterations (v5 to YOLO26) in high-speed aircraft scenarios.
- To simulate the detection-to-intercept feedback loop using a 122mm rocket platform model.
- To evaluate system robustness under different environmental conditions such as occlusion and varying viewpoints.

1.5 Scope and Limitations of the Study

This study covers the implementation of one-step and two-step deep learning detectors for 2D aerial imagery and the mathematical modeling of missile guidance. The research is limited to low-to-medium maneuverability targets and focuses on terrestrial edge-computing hardware. It does not cover the physical manufacturing of hardware prototypes or the analysis of stealth-capable aircraft. Technical constraints include the reliance on the MS COCO and PASCAL VOC datasets for initial training.

1.6 Methodology Overview

The methodology begins with a theoretical study of hierarchical feature representation, followed by the design of a modified YOLO26 architecture. The system is then trained using the MuSGD optimizer and validated through simulations in the MATLAB/Simulink environment. Finally, performance is evaluated based on Mean Average Precision (mAP) and inference latency on CPU and GPU platforms.

1.7 Organization of the Thesis

This thesis is organized into seven chapters. Chapter One introduces the study. Chapter Two provides the theoretical background and literature review. Chapter Three details the system design. Chapter Four explains the simulation environment. Chapter Five presents the hardware implementation. Chapter Six discusses the results, and Chapter Seven concludes the study.

Chapter 2

Theoretical Background and Literature Review

2.1 Part I: Theoretical Background

2.1.1 Fundamental Concepts

Object detection is defined as a dual-mission task consisting of **classification** (identifying what the object is) and **localization** (identifying where the object is). In deep learning, this is achieved through **Convolutional Neural Networks (CNNs)**, which use a hierarchical feature representation.

- **Feature Maps:** 3D matrices representing pixel intensities that are transformed through layers to identify textures, edges, and eventually high-level semantic features like aircraft wings or fuselages.
- **Receptive Field:** The specific portion of the input image that a particular CNN feature is "looking" at.
- **Bounding Boxes:** Defined by coordinates (x, y, w, h), used to finalize the object's spatial location.

2.1.2 Related Systems and Models

The evolution of detection systems is split into two primary paths:

1. **Region Proposal-Based (Two-Step):** Systems like R-CNN and its descendants (Fast and Faster R-CNN) scan for "interesting" regions before classifying them. These are accurate but computationally expensive.
2. **Regression-Based (One-Step):** Systems like YOLO and SSD map pixels straight to coordinates in one pass. These are designed for real-time speed.

2.1.3 Techniques, Methods, or Algorithms

Modern detection relies on several key algorithms:

- **Feature Pyramid Networks (FPN):** These solve the scale variance problem by building a top-down pathway to create high-resolution semantic features at all levels.
- **Non-Maximum Suppression (NMS):** A technique used to filter out redundant, overlapping bounding boxes to keep only the most confident detection.
- **LQR Control:** Used in the guidance phase to execute commands without destabilizing the airframe, optimizing the trade-off between control effort and tracking performance.

2.2 Part II: Literature Review

2.2.1 Review of Previous Studies

Several key studies have shaped the current landscape of aerial detection and air defense:

Zhao et al. (2019), "Object Detection with Deep Learning: A Review," provided a comprehensive analysis of traditional vs. deep methods, noting that CNNs outperformed handcrafted features like HOG and SIFT by bridging the semantic gap through learned representations.

Sapkota and Karkee (2025), "Ultralytics YOLO Evolution," traced the progression of the YOLO family. They highlighted that YOLOv8 introduced decoupled heads to reduce gradient interference, while the newest YOLO26 achieved a 43% faster inference on CPUs by removing Distribution Focal Loss (DFL) and NMS bottlenecks.

Ardiansyah et al. (2025), "Guidance and Control System Design," explored the feasibility of converting 122mm unguided rockets into Surface-to-Air Missiles (SAM). Their results showed that Proportional Navigation (PN) is effective for low-maneuverability targets, achieving miss distances between 0.8 and 2.2 meters.

Malinowski (2020), "Hypersonic Weapon as a New Challenge," discussed the intensification of air threats and the need for automation in Command and Control (C2) systems to handle targets traveling above Mach 5, where human decision-making reaches its limit.

2.2.2 Comparative Analysis of Previous Work

The following table compares the dominant architectures used in aerial target detection:

Table 2.1: Comparison of Detection Architectures

Architecture	**Approach**	**Strengths**	**Weaknesses**
Faster R-CNN	Two-Step	High accuracy, robust	Slow (5 FPS)
YOLOv5	One-Step	Modular, PyTorch native	NMS Bottleneck
SSD300	One-Step	Multi-scale maps	Struggles with small objects
YOLO26	One-Step	NMS-free, Edge-optimized	Newest, requires validation

2.2.3 Research Gap and Project Contribution

While the transition to deep learning has solved many issues related to manual feature extraction, a significant gap remains in the native integration of detection and high-speed guidance for low-power edge devices. Most existing research treats the "detection" and "control" as separate pipelines, leading to latency that is unacceptable for anti-aircraft applications. This project contributes to the field by implementing the NMS-free architecture of YOLO26 specifically for aerial target profiles and integrating it into a closed-loop simulation with LQR-based guidance. This approach aims to prove that a unified, indigenous system can match the performance of high-cost, foreign-sourced air defense clusters.